Energy Science and Engineering

	Normative 4-year Degree
Major	Energy Science and Engineering 能源科學及工程學
Degree / Award Title	Bachelor of Engineering in Energy Science and Engineering 工學士(能源科學及工程學)
Offering Academic Unit	School of Energy and Environment
Normal Period of Study	4 years
Maximum Period of Study	8 years
Credit Units Required for Graduation	125 / eSTAR:129 / INSPIRE:132 credit units

Note: The following curriculum information is subject to periodic review and changes.

GE Requirements (31 credit units) Catalogue Term : Semester A 2025/26

		Credit Units
GE University Requirements	GE1401 University English	3
	GE2410 English for Engineering	3
	GE1501 Chinese Civilisation - History and Philosophy	3
	GE1601 Whole-Person Development	1
GE Distributional Requirements	Take at least 1 course from each of the three distributional areas: Area 1: Arts and Humanities Area 2: Study of Societies, Social and Business Organisations Area 3: Science and Technology	12

	Visit the GE website for more course information.	
School-specified Courses	CA1167 Engineering Communication	3
	SEE1003 Introduction to Sustainable Energy and Environmental Engineering	3
	SEE3002 Energy and Environmental Economics	3
School-specified Courses *(for eSTAR stream)*	CA1167 Engineering Communication	3
	SEE3002 Energy and Environmental Economics	3
	Any course of 3 credit units to be taken in the minor programmes of (i) Computing, (ii) Data Science, (iii) Innovation Management and Entrepreneurship, or (iv) International Business	3

College / School Requirements (18 or 3 credit units)

Catalogue Term : Semester A 2025/26

(i) For taking the major of "Energy Science and Engineering" or "Environmental Science and Engineering" (18 credit units)

Course Code	Course Title	Credit Units
CHEM1200	Discovery in Biology	3
CHEM1300	Principles of General Chemistry	3
PHY1201	General Physics I	3
SEE1000	Professional Development: Career Planning Workshop	0
SEE1002	Introduction to Computing for Energy and Environment	3
SEE2000	Professional Development I	0
SEE4000	Professional Development II	0

Select either MA1200 or MA1300:

Course Code	Course Title	Credit Units
MA1200	Calculus and Basic Linear Algebra I	3
MA1300	Enhanced Calculus and Linear Algebra I	3

Select either MA1201 or MA1301:

Course Code	Course Title	Credit Units
MA1201	Calculus and Basic Linear Algebra II	3
MA1301	Enhanced Calculus and Linear Algebra II	3

(ii) For taking the stream of "Energy/Environment in Science, Technology & Advanced Research (eSTAR)" (3 credits)

Course Code	Course Title	Credit Units
SEE1000	Professional Development: Career Planning Workshop	0
SEE1002	Introduction to Computing for Energy and Environment	3

SEE2000	Professional Development I	0
SEE4000	Professional Development II	0

Major Requirements (no stream/eSTAR:76, INSPIRE:83 credit units)

Catalogue Term : Semester A 2026/27

Notes:

The catalogue term of the major requirement that students will follow will be the effective term of the declared/allocated major.

For normative 4-year degree students who will join the majors allocation exercise, the catalogue term of major requirement will be one year after admission.

For 4-year degree students who already have a major at the time of admission, the catalogue term of major requirement will be the same as their admission term.

1. Basic Core Courses

Normative 4-year Degree / Normative 4-year Degree eSTAR Stream (19 credit units)

Course Code	Course Title	Credit Units
MA2181	Mathematical Methods for Engineering	3
SEE2001	Electromagnetic Principles for Energy Engineers	3
SEE2002	Chemical Sciences for Energy and Environmental Engineers	4
SEE2003	Introduction to Energy and Environmental Data Analysis	3
SEE2101	Engineering Thermofluids I	3
SEE2201	Fundamentals of Environmental Engineering	3

Normative 4-year Degree INSPIRE Stream (26 credit units)

Course Code	Course Title	Credit Units	Remarks
MA2181	Mathematical Methods for Engineering	3	
SEE2001	Electromagnetic Principles for Energy Engineers	3	
SEE2002	Chemical Sciences for Energy and Environmental Engineers	4	
SEE2003	Introduction to Energy and Environmental Data Analysis	3	
SEE2101	Engineering Thermofluids I	3	
SEE2201	Fundamentals of Environmental Engineering	3	
SEE4993	Research Seminar	1	
SEE4994	Guided Study in Energy and Environment	3	*

SEE4995	Guided Study in Energy and Environment	6	*
SEE4998	Special Project in Energy and Environment	3	*
SEE4999	Special Project in Energy and Environment	6	*

**Remark: Select any one among the following three options:*
- SEE4994 + SEE4998
- SEE4995
- SEE4999

2. Major Core Courses (45 credit units)

Course Code	Course Title	Credit Units
SEE3001	Energy and Environmental Policy	3
SEE3003	Climate Change and Adaptation Strategies	3
SEE3101	Engineering Thermofluids II	4
SEE3102	Power Plant Engineering	3
SEE3103	Energy Efficiency for Buildings	3
SEE3104	Sustainable and Renewable Energy	3
SEE4001	Engineers in Society	1
SEE4003	Energy and Environmental Engineering Laboratory	3
SEE4004	Environmental Impact Assessment for Sustainable Development	4
SEE4112	Sustainable Engineering Systems: Modelling and Analysis	3
SEE4216	Combustion and Air Pollution Control	3
SEE4217	Waste and Wastewater Treatment Engineering	3
SEE4997	Final Year Project	6
SYE4024	Project Management	3

3. Electives (12 credit units)

Select at least **three** courses from the following list:

Course Code	Course Title	Credit Units
SDSC3002	Data Mining	3
SEE4113	Nanotechnology in Energy Conversion and Storage: Concepts and Creative Science	3
SEE4114	Bioenergy Engineering: Principles and Applications	3
SEE4115	Energy Catalysis and Reaction Engineering	3
SEE4116	Energy and Carbon Auditing	3
SEE4117	Solar Energy Engineering	3
SEE4118	Wind and Marine Energy	3
SEE4119	Electrical Energy Conversion	3

SEE4120	Materials Engineering for Energy Applications	3
SEE4121	Gas Engineering	3
SEE4122	Chemical Separations for Energy and Environmental Applications	3

Select at least **one** course from the following list:

Course Code	Course Title	Credit Units
SEE3201	Atmospheric Science - An Introductory Survey	3
SEE3204	Urban Sustainability	3
SEE3205	Urban Sustainability	3
SEE3206	Environmental Social Governance	3
SEE3207	Indoor Environmental Quality	3
SEE4202	Atmospheric Chemistry	3
SEE4205	Design of Smart Cities and Sustainable Building	3
SEE4218	Water and Water Resource Engineering	3
SEE4220	Measurements of Air Pollutants	3

4. Optional Electives (15 credit units)

Students may choose to enroll in all of the following course(s) if they are interested in being a member of The Hong Kong Institution of Engineers (HKIE) in the Building Services (BSS) discipline.

Course Code	Course Title	Credit Units
CA3712	Electrical Services	3
CA3722	HVAC Engineering	3
CA3732	Fire Engineering and Piped Services	3
CA4718	Power Electronics and Smart Lighting Controls	3
CA4737	Fire Science and Modelling	3

For taking the stream of "Energy/Environment in Science, Technology & Advanced Research (eSTAR)"

The “eSTAR” stream encompasses (i) ESE major; (ii) a research component, and (iii) minor study in one of the four designated areas, namely Computing, Data Science, Innovation Management and Entrepreneurship, and International Business.

(i) ESE major as mentioned above. (76 credit units)

(ii) Research (7 credit units or above)

a. Guided Study and/or Research (Take any one among the following three options for 2 semesters.)

Course Code	Course Title	Credit Units	Remarks
SEE4994	Guided Study in Energy and Environment	3	Option 1

SEE4998	Special Project in Energy and Environment	3	Option 1
SEE4995	Guided Study in Energy and Environment	6	Option 2
SEE4999	Special Project in Energy and Environment	6	Option 3

b. Research Seminar (Students will attend a minimum of 5 research seminars per semester.)

Course Code	Course Title	Credit Units
SEE4993	Research Seminar	1

c. MSc- or PhD- level courses with the approval of the Major Leader. (Optional)

(iii) Minor (15 credit units or above)

Computing (COMP) – 15 credit units

Data Science (DSC) – 18 credit units

Innovation Management and Entrepreneurship (IME) – 15 credit units

International Business (IB) – 15 credit units

Note:
The following courses (11 credit units in total) must be completed before taking DSC minor core courses: CS2311 Computer Programming, MA2503 Linear Algebra, and MA2506 Probability and Statistics.

Suggested Study Plan

Recommended Study Plans for ESE Students